JKU AI · HACKATHON DEMO

SARGE

The strict study coach that plans backward from your exam, and changes the plan when you get things wrong.

Team mates · Hacktoberfest Hack Day Linz · MLH x Young AI Leaders Linz

sarge-study-coach.vercel.app

Demo with fictional course data

THE PROBLEM

EXAM IN 12 DAYS. 200 SLIDES. NOW WHAT?

- Rereading and highlighting are low-utility study techniques
- It's hard to see what you don't know
- Cramming can feel productive, but retention fades

THE OUTCOME

WALK INTO THE EXAM KNOWING YOU'RE READY

$$\text{Value} = \frac{\text{Dream outcome} \times \text{Likelihood of success}}{\text{Time} \times \text{Effort}}$$

Dream outcome

Ready on exam day, with a rest day before it

Likelihood

Weak topics get extra drills automatically

Time

A full 12-day plan right after onboarding

Effort

Six questions, then Sarge gives the orders

HOW IT WORKS

FROM COURSE SLIDES TO BATTLE PLAN

01 ENLISTMENT

Six onboarding questions

02 UPLOAD

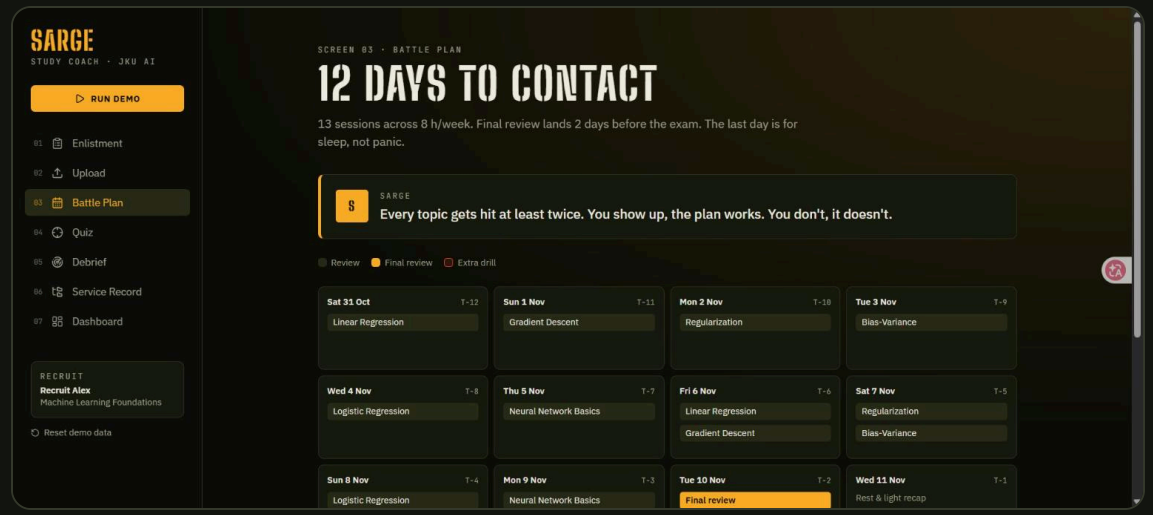
Topic map (simulated in this demo)

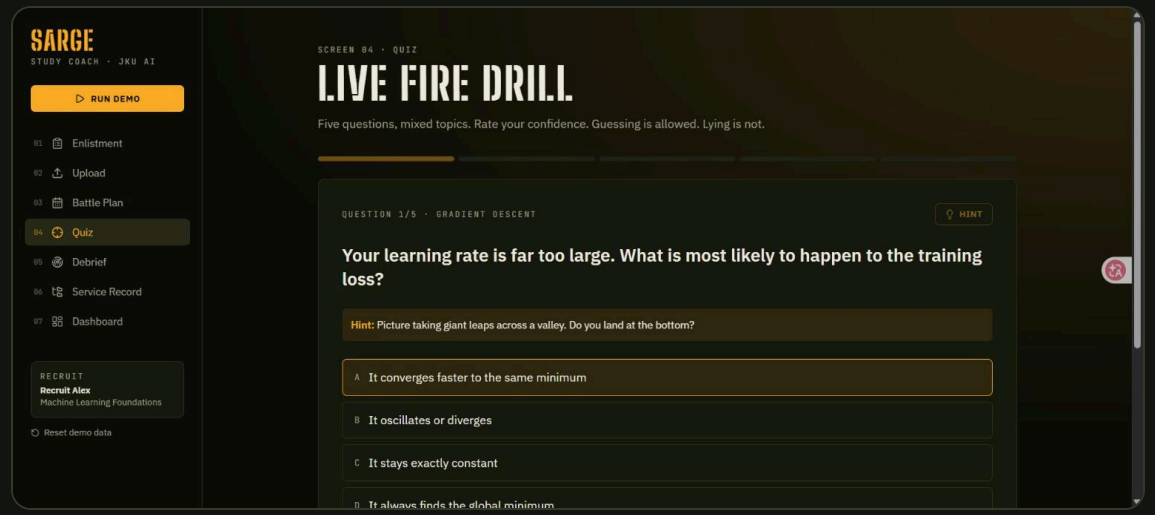
03 BATTLE PLAN

13 sessions, counted back from the exam

04 QUIZ

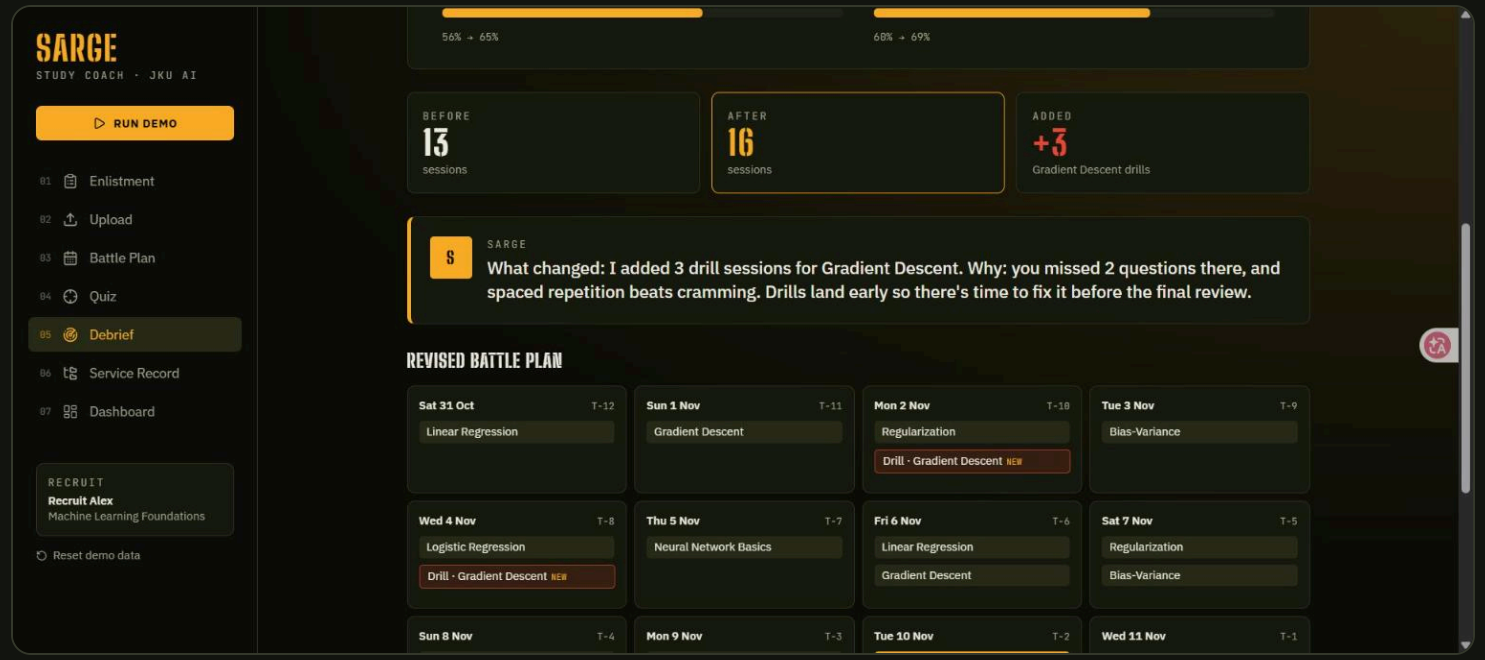
Hints first, plus confidence ratings





THE KEY MOMENT

GET IT WRONG. THE PLAN CHANGES.



13 → 16

sessions, with 3 new drills

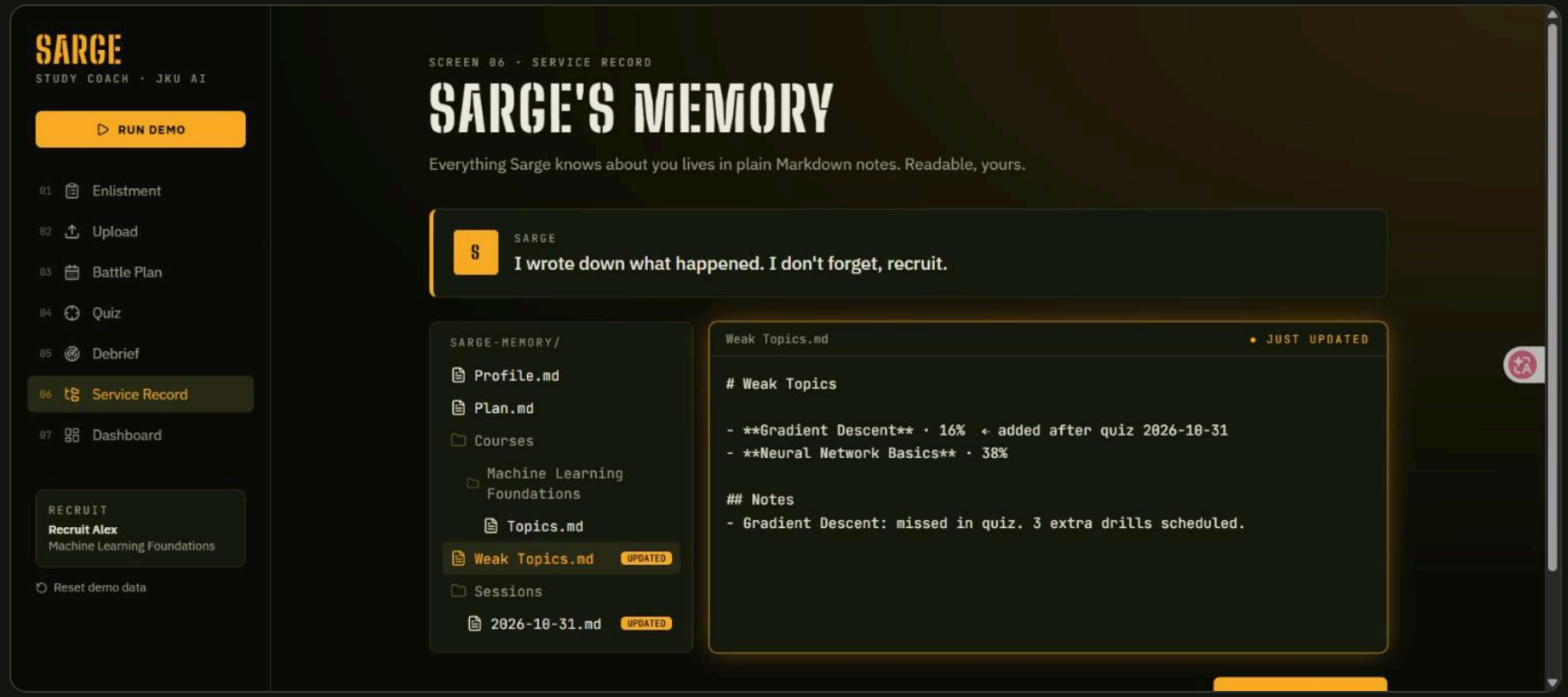
64% → 16%

Gradient Descent mastery after 2 wrong answers

> "Two wrong on gradient descent, recruit. We are not done here."

STUDENT MEMORY

SARGE DOESN'T FORGET



Everything Sarge knows lives in plain Markdown notes the student can read. Weak Topics and the session note update after each quiz.

THE EVIDENCE

WHY THIS DESIGN: LEARNING SCIENCE

Self-testing

Supports learning more than passive review.

In the demo: the quiz

Spaced practice

Supports retention better than cramming.

In the demo: the battle plan

Hints before answers

Unguarded AI can harm later performance without AI.

In the demo: the Hint button

We deliberately don't use "learning styles": evidence does not support matching instruction to a learning style.

THE BUILD

WHAT'S REAL, AND WHAT'S NEXT

BUILT TODAY

Clickable demo with fictional data

Seven screens, with quiz answers driving the plan and memory. Built with Lovable and Claude. MIT licensed.

NEXT

Connect the real brain

Gemma 4 for real slides and quizzes, local Ollama mode, FSRS scheduling, ElevenLabs audio, then KUSSS and Moodle.

What we learned

How to turn learning-science research into a product flow, and how to ship a clickable demo quickly with AI tools.

Study what works, not what feels like work.

sarge-study-coach.vercel.app

github.com/razz-69420/sarge-study-coach